NAME

kh − function

LIBRARY

iapws library (*libiapws*, *−liapws*)

SYNOPSIS

Fortran **pure subroutine kd(***T* **,***gas***,***heavywater***,***k***)**

real(dp), intent(in), contiguous ::*T(:)*

character(len=*), intent(in) :: *gas*

integer(int32), intent(in) :: *heavywater*

real(dp), intent(out), contiguous :: *k(:)*

C **void iapws_g704_kd(double ****T*,**char ****gas*,**int** *heavywater*,**double ****k*,**int** *size_gas*,**size_t** *size_T***)**

Python **kd(***T:***np.ndarray,***gas:***str,***heavywater:***bool=False)->Union[np.ndarray,float]:**

DESCRIPTION

Computes the Henry constant *kH* in MPa for a given temperature *T*. Compute the vapor-liquid constant *kD* for a given temperature *T*.

Arguments:

T(:)	Temperature in K.
gas	Gas.
heavywater	Flag if D2O (1) is used or H2O (0).
k(:)	Vapor-liquid constant in MPa.

RETURN VALUE

k is filled with NaN if gas is not found.

EXAMPLES

Fortran:

```
use iapws
real(dp) :: T(1), kD(1)
T(1) = 25.0_dp + 273.15_dp
call kd(T, "O2", 0, kD)
```

C:

```
#include "iapws.h"
double T = 25.0 + 273.15;
double kD;
iapws_g704_kd(&T, "O2", 0, &kD, strlen(gas), 1);
```

Python:

```
import numpy as np
import pyiapws
```

```
T = np.asarray([25.0])
kD = pyiapws.kd(T+273.15, "O2", False)
print(kD)
```

SEE ALSO

iapws(3) **iapws_cli**(1) **iapws_kh**(3) **ciaaw**(3), **codata**(3),